

Reality Check on Side-Channels: Lessons learnt from breaking AES on ARM Cortex-A72 processor with Out-of-Order Execution

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Abstract

Side-channel analysis (SCA) has posed a significant threat to systems for nearly three decades. Numerous practical demonstrations have targeted everyday devices, such as smart cards, cryptocurrency wallets, and smartphones. However, much of the research in the public domain has focused on low-end microcontrollers, limiting our understanding of the challenges involved in attacking more complex systems. In this work, we conduct a reality check on SCA by targeting a high-performance ARM Cortex-A72 out-of-order processor, commonly found in smartphones. We evaluate the practical effort required for key recovery attacks, considering various threat models, from basic to advanced. Our results show that while basic approaches fail, advanced approaches like deep learning-based SCA can successfully recover the secret key. This multi-tier evaluation approach is crucial for comprehensive risk assessment and informed decision-making regarding mitigation strategies, balancing security, performance, and area constraints.

1 Introduction

As our world becomes increasingly interconnected, microchips are exposed to vast amounts of sensitive data. Securing these chips against sophisticated attacks is important to safeguarding user privacy and security. However, accurately assessing the real-world impact of such attacks remains a complex challenge.

Traditionally, chip design has prioritized area, power, and performance as key design metrics. Recently, security has emerged as the fourth design metric. For instance, modern

Intel processors are hardened against several physical attacks¹. However, these protections were integrated nearly two decades after the initial attacks were identified. This highlights a common trend: *chip manufacturers are often motivated to implement security countermeasures only after a specific threat has been well-understood.*

In this paper, we focus on electromagnetic side-channel attacks (SCA) targeting advanced embedded devices like ARM Cortex-A processors. SCA primarily targets cryptographic operations to recover the underlying secret key, which is essential to the system's security.

While much of the existing research has concentrated on simple, in-order cores like the 8-bit AVR ATMEGA or 32-bit ARM Cortex-M, these devices are often relatively easy to compromise. For example, even a high-order masking scheme (such as fifth-order, significantly exceeding state-of-the-art protection) has been successfully broken on an ARM Cortex-M4 [5]. While this demonstrates the power of SCA, it may not accurately reflect the practical feasibility of such attacks against modern, high-end processors found in everyday devices like smartphones and wearables.

1.1 Related Works

There are only a few works that report practical SCA on ARM Cortex-A processors. Balasch et al [1] and Longo et al [11] demonstrated practical attacks on AES when implemented on an ARM Cortex-A8 processor (BeagleBone Black development board). The attacks required significant effort with measurement filtering and alignment, but were shown successful in different settings including software-only implementation, hardware-coprocessor and a NEON3 core. Similarly, Frieslaar and Irwin [6] reported practical attacks on ARM Cortex-A7 (Raspberry Pi 2). Further, AES-128 was shown exploitable on Raspberry Pi 2B v1.2 with ARM Cortex-A53 [7, 18]. These results were reported with the target underclocked from 1 GHz to 600 MHz. All the attacks required significant effort

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¹<https://www.intel.com/content/www/us/en/newsroom/news/the-story-behind-new-intel-security-feature.html#gs.hh01bz>

Table 1: Comparison of related works on practical side-channel attacks on ARM Cortex-A processors.

	Target Processor	Platform	Adversary Type
Balasch et al. [1]	ARM Cortex-A8	BeagleBone Black	Educated
Longo et al. [11]	ARM Cortex-A8	BeagleBone Black	Educated
Frieslaar and Irwin [6]	ARM Cortex-A7	Raspberry Pi 2	Educated
Wang et al. [18]	ARM Cortex-A53	Raspberry Pi 2B v1.2	Educated
Haas and Aysu [9]	ARM Hurricane (iPhone 7)	iPhone 7	Advanced
Wang et al. [18]	ARM Cortex-A53	Raspberry Pi 2B	Advanced
Current Work	ARM Cortex-A72	Raspberry Pi 4B	Basic to Advanced

in trace filtering and signal realignment. We classify such an adversary as educated who is capable of applying variety of signal processing methods to recover a clean trace. A basic adversary is expected to collect the traces and apply common attacks like correlation power analysis (CPA) to recover the key. An educated adversary will apply CPA after cleaning the traces. At DAC 2022, Haas and Aysu [9] demonstrated first attack on iPhone 7 targeting a standard ARM CE AES implementation. The underlying ARM core is Hurricane running at 2.36 GHz, which is from the same generation as ARM Cortex-A53. The attack can also be classified under an educated to advanced attacker as it required PCB modification, advanced filtering and setup upgraded to make the attack realistic. Wang et al. [18] also report practical attacks on ARM Cortex-A53 considering an advanced adversary who has access to a clone device to profile the leakage under known key setting and exploit the unknown key leakage using deep-learning.

Table 1 summarises the related work. While type of adversary (basic, educated and advanced) is briefly mentioned before, more details can be found in Section 3.

1.2 Motivation and Contribution

Practical side-channel attacks on advanced ARM Cortex-A processors have been demonstrated in prior research. However, the effort required to execute such attacks—considering different adversary capabilities and threat models—remains poorly understood. This gap makes it challenging for designers to accurately assess risk levels, prioritize defenses, and justify the cost of security measures.

In this work, we address this challenge by conducting a comprehensive side-channel analysis on the Raspberry Pi 4B, powered by an out-of-order ARM Cortex-A72 processor. Our goal is not to demonstrate feasibility—this has been shown—but to provide a systematically evaluate the time, resources, and expertise needed by adversaries with varying capabilities. In a nutshell, the key contributions of this work are

1. We conduct a comprehensive side-channel analysis of AES on the Raspberry Pi 4B (ARM Cortex-A72), highlighting challenges like noise, jitter, OS interference, and

out-of-order execution.

2. We propose a multi-level threat model to assess the effort required by adversaries with varying capabilities for key recovery.
3. We present a novel AES dataset collected from a complex processor, offering improved realism and explainability for DL-SCA research. This dataset will be publicly released upon acceptance.

Our findings provide practical insights² to help designers make informed decisions when planning and budgeting for security countermeasures.

2 Experimental Setup and Trace Acquisition

This section presents our experimental setup. It also highlights the key challenges faced during trace collection and proposed workarounds.

2.1 Target Board and Implementations

We target the Raspberry Pi 4B board with a Broadcom BCM2711 SoC. It features a quad-core out-of-order ARM Cortex-A72 (ARMv8, 64-bit) processor, operating at a maximum frequency of 1.8 GHz. This device lacks hardware support for AES. It runs on the Bullseye version of the Raspberry Pi OS (Debian Linux). We execute a C-based implementation of AES-128 for our experiments without computer optimisations. Three different targets are considered: standalone AES S-box, complete AES-128 encryption (not hardened by additional countermeasures) and AES S-box hardened by first-order Boolean masking. Notably, this is the first exploration of side-channel vulnerabilities in an out-of-order (OoO) processing environment.

²The relevant artifacts have been made public at <https://doi.org/10.5281/zenodo.15524301>

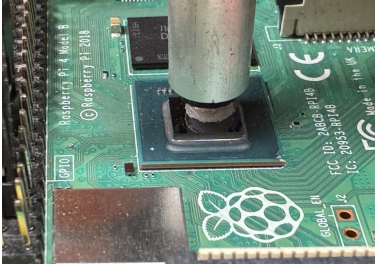


Figure 1: EM probe positioned on the ARM Cortex A72

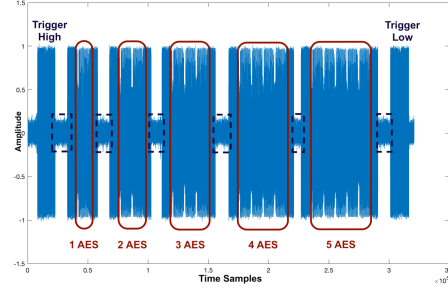


Figure 2: Stable EM trace

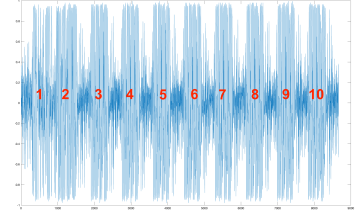


Figure 3: AES 10 rounds visible in a stable EM trace

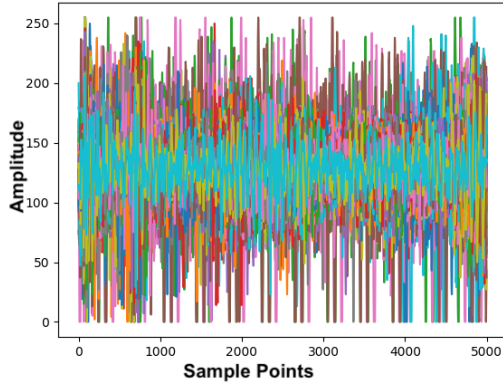


Figure 4: Plot of 200 traces of AES S-Box

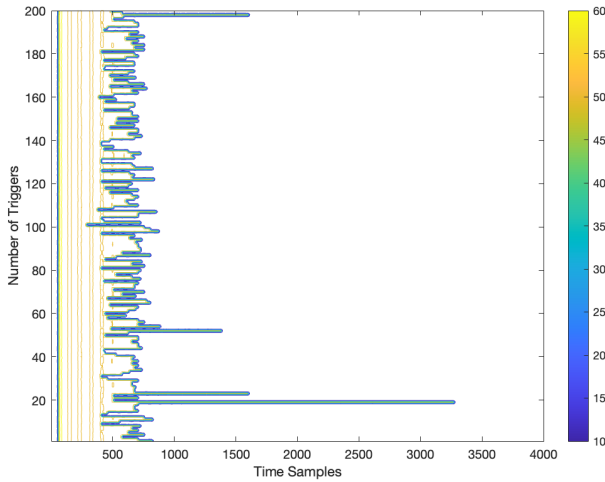


Figure 5: Contour plot of 200 trigger length of S-box traces.

2.2 Trace Collection

For side-channel trace acquisition, we use a Riscure ZVC45 series EM probe with an integrated amplifier, connected to a LeCroy oscilloscope for data capture. The trigger is synchronised to the start of the operation, with interrupts affecting the background processes. However, it is not trivial to identify

where to place the EM probe.

Initial tests involved collecting electromagnetic (EM) traces by placing the probe near the decoupling capacitors and on the chip surface, while the chip cover remained intact. We began by executing an unprotected AES implementation in C on the Raspberry Pi, systematically moving the EM probe across the board. However, we were unable to observe the expected 10 AES rounds. Despite collecting traces from various locations and performing correlation power analysis (CPA), the results were inconclusive. Even a test vector leakage assessment (TVLA) [8] failed to reveal any significant leakage, leaving uncertainty as to whether the traces were excessively noisy or if the measurement positions were simply not leaking.

To address these challenges, we carefully removed the metal casing from the chip, ensuring no damage was incurred. Figure 1 illustrates the resulting setup, with the probe positioned directly above the chip's surface. To identify a good position for the probe over the chip, we run a sequence of operations with an increasing number of AES executions, starting with one AES followed by a sleep, then two AES executions and a sleep, continuing up to five AES executions, and ending with a final sleep³. As expected, the measurements showed a lot of noise and jitter, but we were able to identify the repeating AES pattern (see Figure 2). It also shows that the AES executions (marked inside red boxes) were sandwiched between context-switching from the OS background processes like sleep (marked inside dashed black boxes). We can observe that the signal is high before and after the AES executions, which is unaccounted for by our implementation. These sample points may stem from the context-switching activity performed by the OS running in the background. This happens especially during the sleep operation, when the processor is idle and is assigned to another background operation, causing a spike in the signal. Finally, we observed AES block with 10 rounds visible as seen in Figure 3, confirming a favourable position for our analysis.

Although the traces shown in Figures 2 and 3 look promising, we note that we have chosen the most stable trace among poor-quality traces. Due to jitter and noise, it is difficult to constantly get good-quality traces despite finding a favourable

³sleep..AES..sleep..AESAES..sleep..AESAESAES..sleep.. and so on

measurement spot. This is further aggravated by the multi-core nature of ARM Cortex A-72, where AES can run in parallel with other background processes as well. Out-of-order execution makes things even worse by further diminishing the information leakage of AES traces. Figure 4 shows a plot of 200 overlapping traces corresponding to AES S-box execution (preceded with a short sleep) with random inputs. The misalignment is evident, highlighting the non-constant time behaviour partly due to out-of-order processing as shown in Figure 5. It is the contour plot of 200 trigger signals enclosing the S-box operation. It can be seen that S-box operation takes 300 to 3500 sample points, which can be seen in blue colour, noting the end of operation. This fluctuation in the time taken causes significant misalignment in the traces.

As a result, we have to put additional effort into aligning the traces. While several methods in the literature are proposed for realigning the traces, there is no standard or recommendation. Generally, evaluators are required to find a good alignment technique based on the target implementation. Therefore, we conduct our evaluation progressively, by performing initial analysis on S-box execution and eventually performing key recovery from full AES. We collected 3 million traces for unprotected and first-order masked AES S-box implementation. For full AES-128 encryption, we collected 4 million traces focusing only on the first round sample points. For clarity and conciseness, those three distinct traces are denoted as *S-trace* (unprotected S-box execution only), *M-trace* (first-order masked S-box execution only) and *F-trace* (full AES encryption). The evaluation of these traces under a multi-level threat model is reported in the next section.

3 Evaluation Methodology & Metrics

In this section, we present the proposed threat models followed by comprehensive evaluation of the target under these threat models.

3.1 Threat Model

We consider an adversary who queries the target device running a cryptography operation with known inputs and aims to recover the secret key. During the operation, the adversary can capture electromagnetic side-channel activity using a near-field antenna. The trace capture is synchronized with the start of the target operation. Inspired from [4], we consider three types of adversaries with increasing levels of expertise.

- **Basic adversary** has access only to a target device where the secret key is fixed and unknown. We assume this adversary is unaware of the implemented countermeasure and thus cannot cancel its effect. As a result, the adversary can only perform non-profiled (or unsupervised) attacks with a priori leakage model like Hamming weight or distance.

- **Educated adversary** has all the characteristics of a basic adversary. Additionally, the educated adversary has partial knowledge of the underlying countermeasure and can use available methods to bypass it. For instance, if a jitter-based countermeasure is present, this adversary will put efforts to align the measurements.
- **Advanced adversary** considers a strong setting where the adversary has access to a clone device. The clone device is queried in a white box setting (known key, input, countermeasure) to build a profiled (or supervised) model. Adversary may use attacks like template or deep learning to build a precise model of the leakage and exploit it to match measurements from the target device to recover the unknown key. Additionally, this adversary has capabilities to bypass countermeasures (e.g., aligning the traces).

Note that in [4] only the Advanced adversary had access to real measurement. In our case, we adapt the threat model in which the adversary at each level works with real measurement. In addition, we perform a comprehensive evaluation using this multilevel adversary, providing designers with a basis for risk analysis.

3.2 Attack Methods & Metrics

Since the divide-and-conquer approach allows one to attack each key byte independently, we concentrate on the first key byte without loss of generality, and thus the attack target is the first output byte of the S-box in the first AES round. During the evaluation, different side-channel metrics are considered. To assess the quality of the measurements, before or after alignment, *Test Vector Leakage Assessment* (TVLA) [8] and *Signal-to-Noise Ratio* (SNR) [13] are applied. *Correlation Power Analysis*, *Key Rank* and *Guessing Entropy* [15] are leveraged to indicate the attack effectiveness. The first set of investigations is performed on a standalone AES S-box (unprotected) and later extended to full AES encryption (unprotected) and masked AES encryption.

3.2.1 Test Vector Leakage Assessment (TVLA)

TVLA [8] involves the computation of univariate Welch's *t*-test over two given sets of side-channel measurements to identify if there are differentiating features between them. Assuming a null hypothesis that the mean of two sets is identical, TVLA will return a decision on whether to accept (*PASS*) the null hypothesis or not (*FAIL*). The formulation of TVLA or the *t*-score over two sets of measurements T and $\tilde{T}$ is given by:

$$s = \frac{\mu_T - \mu_{\tilde{T}}}{\sqrt{\frac{\sigma_T^2}{n_T} + \frac{\sigma_{\tilde{T}}^2}{n_{\tilde{T}}}}}. \quad (1)$$

where, μ_T , σ_T and n_T (respectively $\mu_{\tilde{T}}$, $\sigma_{\tilde{T}}$ and $n_{\tilde{T}}$) are mean, standard deviation and cardinality of the trace set T (respectively $\tilde{T}$). The null hypothesis is rejected with a confidence of 99.9999% only if the absolute value of the t -test score is greater than 4.5, which implies that the two trace/data sets are different and might leak some side-channel information and hence is considered a *FAIL* test.

3.2.2 Signal-to-Noise Ratio (SNR)

SNR defines the ratio between the signal variance and the noise variance. Assuming the leakage depends on values p , the leakage can be estimated as function f of value p and secret $k \in K$, denoted as $L = f(p, k) + N$, where L is the leakage and N is estimated to be independent Gaussian noise. The SNR is computed as:

$$SNR = \frac{V(E[L|X])}{E(V[L|X])}. \quad (2)$$

where, E is expectations and V is variance.

The higher the SNR value, the better the quality of the measurements, and vice versa. As such, as an attacker, the aim is to get measurements with as high SNR as possible.

3.2.3 Correlation Power Analysis (CPA)

One of the most common non-profiling attacks is CPA [2]. It is an effective non-profiled attack that is proven to be optimal [10] if the side-channel measurements linearly depend on the hypothetical leakages. It is done by calculating the Pearson correlation to measure the linear relation between the intermediate values, computed from the hypotheses with different secrets, with the actual leakages. The attacker computes the intermediate values as a function f of known inputs p and (hypothetical) secret $k \in K$. In this case, the attacker will compute $H = f(p, k)$. These intermediate values will then be compared with the actual leakage traces T obtained while processing the actual secret k^* . The secret k with the highest (absolute) correlation can then be estimated as the secret value.

The Pearson correlation between x and y can be computed as follow:

$$r(x, y) = \frac{\sum_{i=1}^N ((x_i - \bar{x})(y_i - \bar{y}))}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}. \quad (3)$$

where, $x = \{x_i\}_{i=0, \dots, N-1}$, $y = \{y_i\}_{i=0, \dots, N-1}$, and $\bar{x}$ and $\bar{y}$ are the average for x and y respectively.

The intermediate values themselves have to be first mapped to a leakage value. In general, a leakage model is used to approximate the behaviour of the measured traces. For the software implementation, the leakage is usually assumed to follow the Hamming weight (HW) model. In contrast, for hardware implementation, it is the Hamming distance or HD

model. In addition to the mentioned leakage values, an identity mapping (ID) can also be used as an alternative leakage model.

3.2.4 Key Rank and Guessing Entropy (GE)

Once the attack has been conducted, the key is sorted or ranked based on the maximum likelihood metric. The top-ranked key will be considered the most likely correct key. The guessing entropy or GE is defined as the average rank of the correct key k over multiple experiments [15]. If $GE = 1$, when using N attack traces, it means that the correct key is ranked first and thus, the attack is considered successful. We denote $NTGE$ to be the least number of traces required to attain $GE = 1$. In the following, GE is obtained by averaging over 100 independent experiments.

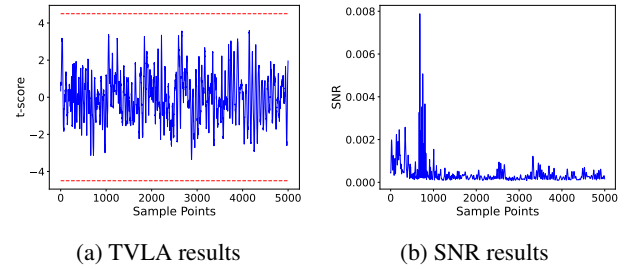


Figure 6: TVLA and SNR results on raw S-trace.

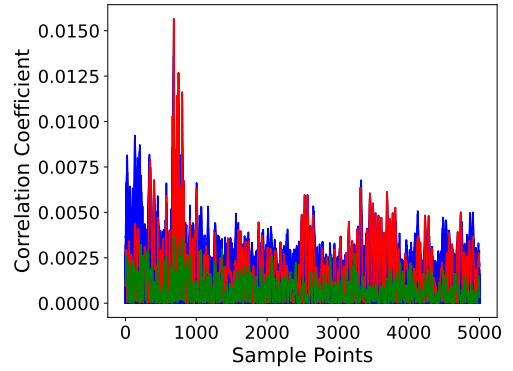


Figure 7: CPA results of 256 key candidates on raw S-trace.

4 Evaluation Results

In this section, we report the comprehensive evaluation of the target traces under a multi-level adversary model.

4.1 Evaluation Under a Basic Adversary

Considering a basic adversary, we first start with leakage identification. TVLA and SNR results computed with all the

3 million raw measurements are shown in Figures 6a and 6b.

Note that the red dashed lines in Figure 6a mark the ± 4.5 threshold. TVLA does not show any leakage as all t -test scores s satisfy $|s| \leq 4.5$. This is expected as the target ARM Cortex-A72 has huge misalignment, and thus univariate techniques like TVLA and SNR are not optimal. SNR, which is computed with all the 3 million raw measurements, shows a small peak observed in the sample points range [600, 1000] and thus suggests potential information leakage. However, the SNR are very low (lower than 0.008), hence it remains unknown whether this possible leakage pertains to the true key and if adversaries can effectively leverage it. We do attempt to recover the key using CPA. The CPA peak as shown in Figure 7 is consistent with the SNR peaks. However, the correct key (shown in green) is not the guessed key (shown in red). The rank of the key with all 3 million traces used for the attacks remains at 200, rendering the attack unsuccessful.

4.2 Evaluation Under an Educated Adversary

In the present case, an *Educated adversary* is aware of the misalignment in the measured traces and would apply methods to overcome it. However, there is no standard approach to overcome misalignment. In this work, we adopt two distinct approaches: *filtering* and *Elastic Alignment* (EA) [17] for alignment.

Filtering is customized for our experimental case. As shown in Figure 5, the duration of the operation varies significantly, leading to substantial desynchronization between the measurements. The applied *filtering* approach checks the distribution of the operation length and selects the ones close to its statistical mode. Thus, a large number of traces are discarded leaving the adversary with 8%-18% of all raw traces. In contrast, EA is more versatile and broadly applicable as a non-profiled approach. EA applies Fast DTW (Dynamic Time Warping) technology for non-linear resampling and thus matches trace segments at varying offsets, making it a robust choice extensively adopted in both academia and also supported in many commercial tools. Previously, Wang et al. [18] demonstrated practical attacks on ARM Cortex-A53 (Raspberry Pi 2B+) using EA as an alignment approach.

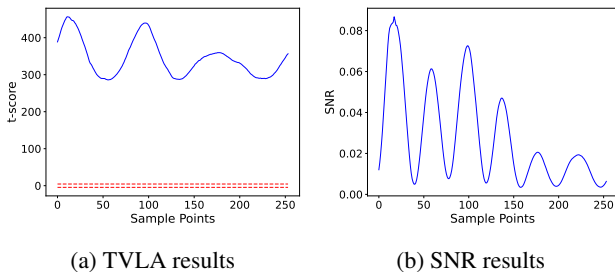


Figure 8: TVLA and SNR results on filtered S-trace.

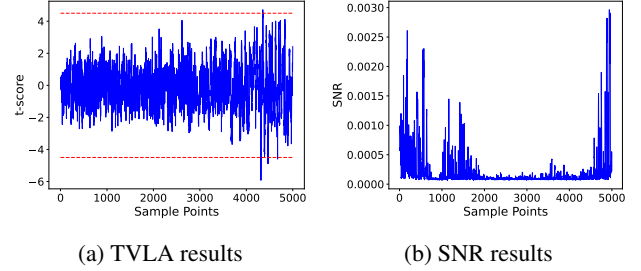


Figure 9: TVLA and SNR results on elastic-aligned S-trace.

After performing *filtering* and EA on S-trace with 3 million measurements, an initial assessment for trace quality is conducted as well. We should highlight that the amount of filtered traces has been decreased to 228,646 from the original 3 million. Besides, the trigger length with highest probabilities has 254 points and thus the points of filtered traces are cut down accordingly. TVLA and SNR results of filtered (resp., elastic-aligned) S-trace are illustrated in Figure 8 (resp., Figure 9). Note that the traces amount for EA is 200,000, ensuring direct comparability with the *filtering* approach. Comparing TVLA results with the raw one in Figure 6a: the t -score results (in Figure 8a) of the filtered traces are significantly higher than the threshold, indicating a substantial amount of leakage across the entire span of points; while it can be observed in Figure 9a that some t -test scores s exceed the threshold ± 4.5 , showing evidence of sensitive leakage in the case of elastic-aligned traces. Considering SNR results, the value of filtered traces (in Figure 8b) increase by 10 times compared to the raw one in Figure 6b. Additionally, SNR results in Figure 9b show improvement after EA, as more leakage peaks are present and the contrast between peak points and flat regions is significantly greater.

After pre-processing, we attempt key recovery using CPA. Figure 10 presents the CPA results for both pre-processing techniques, where red, green and blue lines correspond to the guessed key (by the adversary), correct key and other key candidates, respectively. Again CPA peaks align with SNR peaks. The attacks remain unsuccessful with key ranks of 242 and 65 for *filtering* and EA respectively. The key rank for filtering is worse than the basic attack likely due to reduced available traces. While EA was not able to align the traces well enough for key recovery, other signal processing methods could be explored by the attacker. However, the choice of methods is heavily target dependent and no generic methodologies are available to help an attacker choose the best alignment methods. Thus, the success of such approaches are directly linked to attacker's expertise.

Previously, Wang et al. [18] reported successful CPA on ARM Cortex-A53 with 80,000 traces under the educated adversary. However, as shown in [4] simply moving from in-order to out-of-order core in an ideal noise-free simulation environment can increase the complexity by $3.5\times$. They stated

that this gap is expected to increase several folds when dealing with real noisy measurements. Factoring in the difference in manufacturing technology node of Raspberry Pi 2B+ vs Pi 4B should contribute to an increase in the difficulty of the attack.

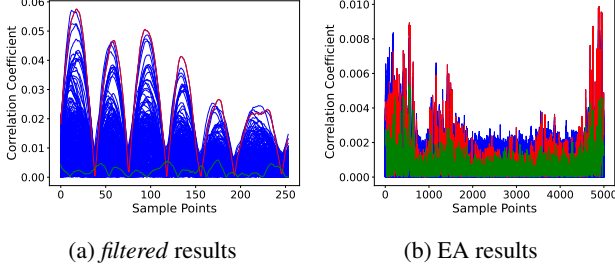


Figure 10: CPA Results under the Educated Adversary

Table 2: Range for hyperparameter random search.

Parameters	Range
Batch Size	100, 200, ... 1000
Learning Rate	1e-3, 5e-4, 1e-4, 5e-5, 1e-5
Optimizer	RMSprop, Adam
Layers	1, 2, ..., 8
Neurons	10, 20, 50, 100, 200, 300, 400, 500
Kernel Initializers	{random, glorot, he}_uniform
Pooling	{max, average}_pool
Pooling size	2, 4, 6, 8, 10
Convolution layer	1, 2, 3, 4
Filters	4, 8, 12, 16
Kernels	26, 28, ..., 52
Padding	0, 4, ..., 16

4.3 Evaluation Under an Advanced adversary

An *advanced adversary* assumes full access to the clone device, enabling profiled/supervised attacks. Recent works [3, 12, 16] show that Deep Learning (DL)-based approaches can surpass classical profiling-based attacks like template attacks in practical setting (high noise, limited traces, countermeasures). Consequently, we choose to leverage Convolutional Neural Networks (CNNs), a commonly used learning paradigm in DL. The advantages of CNNs for SCA are twofold. On the one hand, CNNs essentially address jitters and misalignment in traces (which are major contributors to failed attacks in *basic evaluation* and *educated evaluation*) due to its shift-invariant properties [3]. On the other hand, CNNs are enhanced to handle masked implementations, as they naturally combine samples involving masking shares during training [12].

CNNs require tuning of numerous hyperparameters (which significantly impact the performance), including the number

of layers, the kernel size, the activation function type, etc. To optimize the hyperparameters, we perform a basic random search over the parameter space, which has been used frequently in previous works [14, 19, 20]. In our case, we conduct 100 trials, where each trial involves training a CNN with randomly selected hyperparameters. Specifically in each trial, the model performance is assessed on the validation dataset to improve hyperparameters, followed by evaluation on the test dataset. Table 2 provides the parameter space for hyperparameter search. Note that our goal in this work is not to identify the most optimal model but to recover the key. In fact, formal approaches to find an optimal model remain an open research problem in SCA.

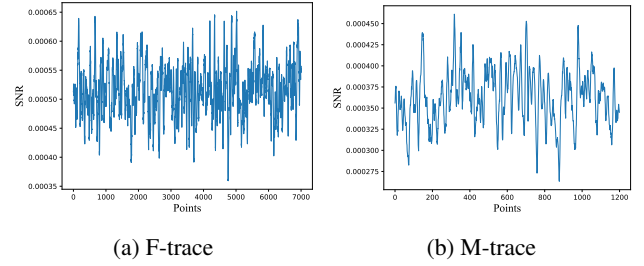


Figure 11: SNR results of F-trace and M-trace.

The raw *S-trace* measurement quality has already been evaluated in Section 4.1. Figure 11a and Figure 11b shows the measurement quality of F-trace and M-trace, which depicts low SNR for an attacker to exploit. Given that an *Advanced* adversary has comprehensive knowledge of the target device and robust pre-processing capabilities, we reasonably assume that it can effectively focus its trace samples exclusively on the S-box operations. Hence for both F-trace and M-trace, the amounts of sample points is reduced to 7,000 and 1200, respectively. It is indicated in both Figure 11a and Figure 11b that SNR shows little leakage of sensitive variables on F-trace and M-trace. Accordingly, two pre-processing techniques, *filtering* and EA (similar to *educated evaluation*), are applied to improve the attack effectiveness for all three datasets i.e. S-trace, F-trace and M-trace.

As shown in Table 3, *advanced* adversary has the capability to retrieve the true key from noisy and misaligned measurements, even without any pre-processing. Precisely, the key can be recovered from raw S-trace with 30,000 traces, when 100,000 traces are used for training the model. Although raw F-trace fails in the same setting, the *advanced adversary* has the capability to directly extract the key from raw M-traces—CNNs decrease GE to 1 using a minimum of 197,578 traces, with 4 million (resp., 100,000) traces for training (resp., attack) phase.

When applying pre-processing techniques, *filtering* and EA reduce the measurements required for a successful attack on raw S-trace to 6,132 and 3,902, respectively, from 23,366 in the same attack settings. For F-trace, the *filtering*

Table 3: Evaluation under an Advanced Adversary. The number of features, the number of traces required for training and the number of traces required for attack are represented as “No of Feats”, “No of Train”, “No of Att”. “NTGE” represents the least number of traces required to attain GE as 1.

Dataset	No of Feats.	No of Train	No of Att	NTGE
Raw <i>S-trace</i>	5,000	400,000	30,000	23,366
Filtered <i>S-trace</i>	405	100,000	30,000	6,132
EA <i>S-trace</i>	500	100,000	30,000	3,902
Raw <i>F-trace</i>	7,000	150,000	50,000	49,799
Filtered <i>F-trace</i>	7,000	100,000	30,000	21,626
EA <i>F-trace</i>	7,000	150,000	50,000	49,989
Raw <i>M-trace</i>	1200	400,000	200,000	197,578
Filtered <i>M-trace</i>	624	400,000	200,000	92,670
EA <i>M-trace</i>	1,000	400,000	200,000	193,961

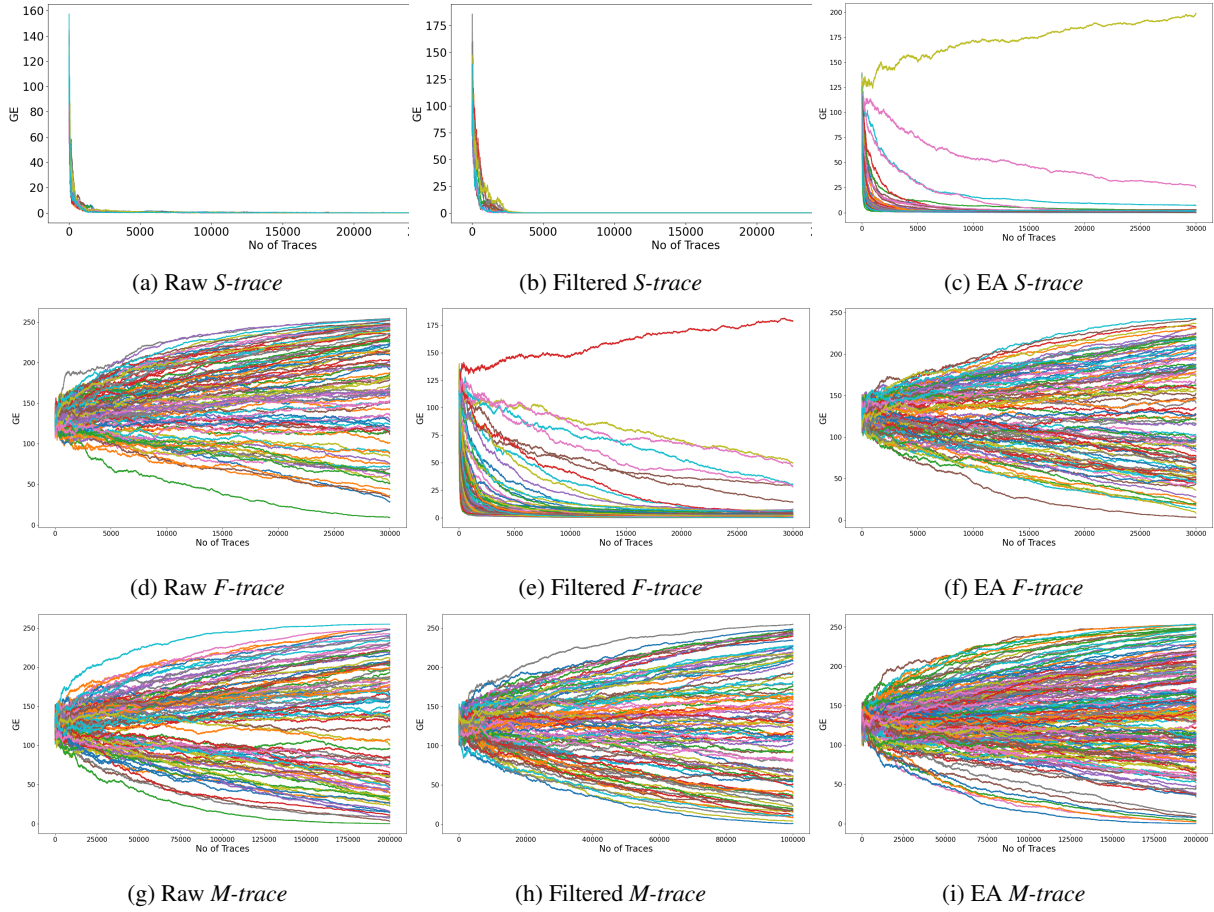


Figure 12: Guessing Entropy analysis for 100 Random Search experiments on different datasets. We perform this analysis on a smaller number of traces.

technique enables successful key recovery with fewer traces, representing a significant enhancement over key recovery attempts observed with raw and EA F-traces. This also shows that the filtering method works better over EA for F-traces. We observe a similar trend for M-trace, where with 400,000

training traces, *filtering* and EA approaches recover the key in 92,670 and 193,961, respectively. Note that raw M-trace required $10\times$ training traces for a successful attack. *Filtering* outperforms EA on both M-traces and F-traces. EA demonstrates limited stability as a preprocessing technique, as its

effectiveness is significantly influenced by multiple factors, including the selection of reference trace and the length of targeted traces. Furthermore, in scenarios involving masked implementations, EA is typically effective at aligning a single point but struggles to align multiple points simultaneously.

To further analyze the capabilities of the *advanced adversary* on multiple trace sets with different pre-processing techniques, the GE results of all the models in 100 random search experiments for S-trace, F-trace and M-trace are presented in Figure 12. We choose only 30,000 attack traces for GE analysis, to make it consistent with the filtered F-trace result. In general, we can see that F-trace and M-trace are harder to learn compared to S-trace, which is expected. We can also observe how preprocessing improves the learnability of the traces. Further, one good model in each case is enough for successful key recovery.

Finally, all the previous experiments show that advanced modern devices exhibit certain security against both *basic* and *educated* adversaries thanks to their complicated architecture (that thus introduces noise and misalignment). However, even protected implementations remain vulnerable to an *advanced adversary*, arousing the need for tailored and careful design of countermeasures. A risk assessment based approach can help to define the correct threat model for the target device and thus allow a fair vulnerability assessment and protection budgeting.

4.4 Performance Evaluation

An important parameter to determine the effectiveness of the attack is the effort required to do it. Table 4 measures this effort in approximate time for processing 100,000 F-trace, including acquisition, pre-processing and attack stages, for various levels of adversaries. It gives a preliminary understanding of the approximate cost required for a side-channel attack on an advanced embedded device. As different preprocessing methods may require different times, so we compute the Total using the maximum of both techniques. For Advanced Attack, the time includes training for 100 random models. While Advanced adversary requires up to 2 days for successful key recovery, it has a strong requirement for a clone device. It is difficult for a Basic adversary to succeed, without it. Therefore, it is necessary to perform risk assessment in the presence of various levels of adversaries, before countermeasure development.

5 Conclusion

This work focuses on evaluating the effort required for side-channel attacks from AES on the ARM Cortex-A72 processor, using the Raspberry Pi 4B with a full OS running. We found that key recovery on the A72 is more challenging than on simpler cores, due to the out-of-order execution and low-quality

Table 4: Time taken (in hours) by different adversaries for 100,000 F-traces.

Processing	<i>Basic</i>	<i>Educated</i>	<i>Advanced</i>
Collection	3	3	3
<i>filtering</i>	-	1	1
EA	-	1	1
Attack	0.2	0.2	31
Total	3.2	4.2	35
unprotected	×	×	✓
protected	×	×	✓

traces caused by misalignment and low SNR. Our evaluation, under a multi-level threat model, shows that basic and educated adversaries, without prior knowledge or advanced profiling, fail to recover the key. In contrast, an advanced adversary using DL-SCA on a cloned device successfully recovers the key, highlighting the potential of such attacks.

Importantly, we demonstrate that DL-SCA can succeed where traditional CPA fails, even with traces that are difficult to process. Our findings emphasize the complexity of attacks on modern processors like the Cortex-A72, indicating that more research is needed to guide the effective allocation of resources for side-channel countermeasures.

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